

NoiseModelling overview

Pierre Aumond

NoiseModelling Days
September 24, 2026

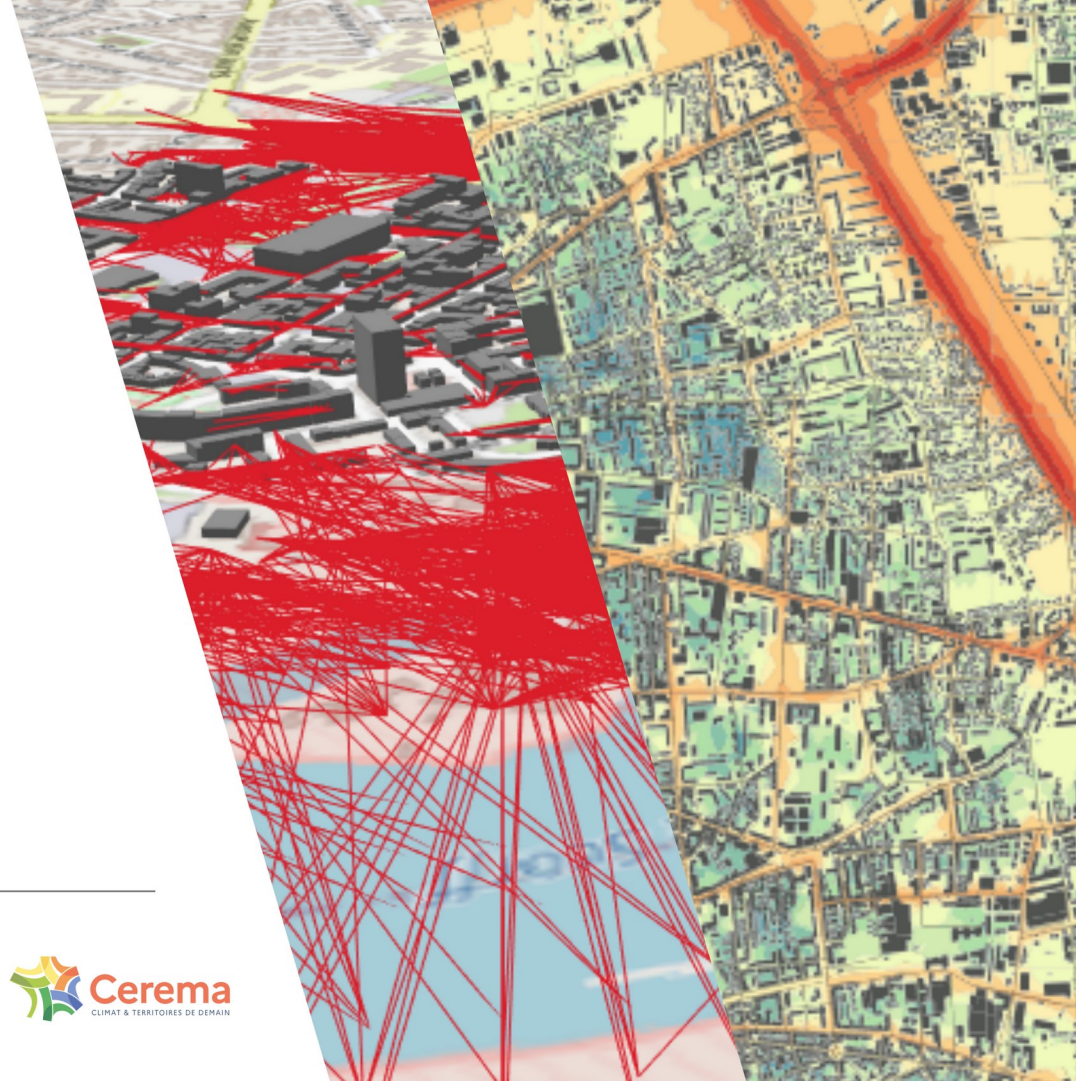
Organized by



Université
Gustave Eiffel



Cerema
CLIMAT & TERRITOIRES DE DEMAIN



Welcome...

... to the NoiseModelling Days 2026 !

NoiseModelling 2026 - Crewmembers



Pierre

Acoustician & Developer

NoiseModelling coordinator

Steering Committee



Gwendall

Geomagician

User side /

documentation

Steering Committee



Martin

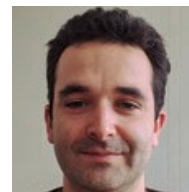
Acoustician / Developer

Harmonoise model
integration



Maguette

Software
Developer



Nicolas

Main Dev.

Software Developer

Steering Committee

6th NM Days

NoiseModelling Days 2026



September 24th 2026

French, UTC+2	Type	Title	Speaker(s)	Affiliation(s)
10h00 - 10h15		Welcome and get to know you better		
10h15 - 10h25	Presentation	NoiseModelling (NM) overview	Pierre Aumond	UMRAE, Univ Gustave Eiffel, Cerema
10h25 - 10h55	Presentation	NoiseModelling News	Pierre Aumond	UMRAE, Univ Gustave Eiffel, Cerema
10h55 - 11h10	Presentation	NoiseModelling - Project tracking page	Gwendall Petit	UMRAE, Univ Gustave Eiffel, Cerema
11h10 - 11h25	Presentation	NoiseModelling - Governance	Pierre Aumond	UMRAE, Univ Gustave Eiffel, Cerema
11h25 - 11h30		Break		
11h30 - 11h45	User presentation	Noise Prediction along Bridges using CNOSSOS-EU and ASJ RTN Models	Junta Tagusari & Seigo Hata	Yokohama City University
11h45 - 12h00	User presentation	A Coupled MATSim-NoiseModelling-Cesium Framework for Three-Dimensional, Multi-Period Simulation and Assessment of Traffic Control Noise Impacts	Yueying Zhang	South China University of Technology
12h00 - 13h30		Lunch break		
13h30 - 14h20	Room 1 - Workshop	For new comers : Discover NoiseModelling in actions (live demo of possibilities)	NM Team	UMRAE, Univ Gustave Eiffel, Cerema
	Room 2 - Discussion	For those who are already familiar with NM : Discussion about the software's shortcomings and the Christmas wish list		
14h20 - 14h30		Break		
14h30 - 14h45	User presentation	NoiseModelling in the Netherlands	Rob van Loon	RIVM
14h15 - 15h00	User presentation	Application of noise modeling for large-scale maps of roads, railways, and wind turbines in Spain at the Barcelona Supercomputing Center	Ignacio Soto-Molina	Ministry for the Ecological Transition of Spain
15h00 - 15h15	User presentation	Noise Modelling in Quito - Ecuador	Paul Lozada	LAMBDA Acústica Y Vibraciones
15h15 - 15h30	User presentation	Noise Mapping in Port Environments: Experimental and Numerical Aspects and Calculations of Exposure Indicators	Matthys Hynek & Olivier Robin	Université de Sherbrooke
15h30 - 16h00	Discussion	Closing and free discussion		



2026

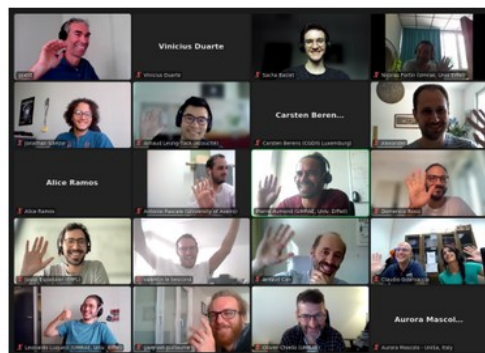
2024 - Internoise

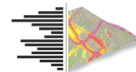
NoiseModelling Days 2025



Day 1 (June 26th 2025)	Time	Topic	Event organizer	Affiliation(s)
09:00 - 09:15	Registration			
09:15 - 10:00	Presentation	Introduction	NoiseModelling Team	
10:00 - 10:15	Presentation	Open Research in acoustics : The Noise Planet & LHM experience	Johel Poirier	UMRAE, Univ Gustave Eiffel, Cerema
10:15 - 10:30	Presentation	Research collaboration open projects inclusion software community	Marcus Winkler	Ernst-Reuter University of Technology
10:30 - 10:45	Break			
10:45 - 11:00	Presentation	Open source data & software for noise and vibration modelling in the Netherlands	Arno van der Vliet	Netherlands Institute for Public Health and the Environment
11:00 - 11:15	Presentation	Using the NoiseModelling software for quantifying industrial noise in the environment	Thomas Huet	CEA
11:15 - 11:30	Presentation	CNOSSOS-EU & Open source	Pierre Aumond	UMRAE, Univ Gustave Eiffel, Cerema
11:30 - 11:45		Break		
11:45 - 12:00		Break		
12:00 - 12:15		Break		
12:15 - 12:30		Break		
12:30 - 12:45		Break		
12:45 - 13:00		Break		
13:00 - 13:15		Break		
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23:30 - 23:45		Break		
23:45 - 00:00		Break		

2025





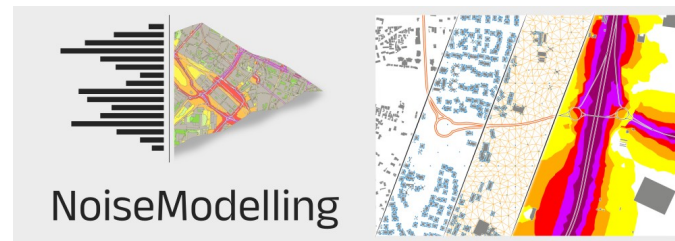
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10h55 - 11h15	Presentation	NoiseModelling - Project tracking and Governance	Gwendall Petit	UMRAE, Univ Gustave Eiffel, Cerema
11h15 - 11h25	Presentation	NoiseModelling in videos	Alexandra L. Montenegro	Università di Pisa, ARPAT 
11h25 - 11h30		Break		
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NoiseModelling in 5 minutes

- Since 2010
- Implement CNOSSOS-EU method
- Road, railway, and “industrial” sound sources
- From local to national scale
- By and “mainly” for research



Public Money

Public Code

Open-Source tool

Open-science initiative
(GPLv3)

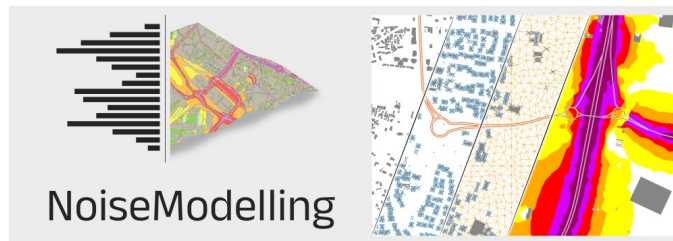
- reproducibility
- transparency
- traceability

Allowed :

- Commercial use
- Modification
- Redistribution
- Selling services based on the software
- Using in your own company

Not allowed :

- Closing the source
- Adding extra license restrictions
- Mixing with incompatible licenses
- Distributing without full source code
- Preventing user from modifying on devices (anti-Tivoization)



Github distributed

github.com/Universite-Gustave-Eiffel/NoiseModelling/

The screenshot shows the GitHub repository for NoiseModelling. At the top, it indicates the repository is public and has 96 forks and 234 stars. The main content area lists files and their commit history. The README section is visible, featuring a map visualization and the text 'NoiseModelling' and 'NoiseModelling is a free and open-source Java library for producing environmental noise maps from local to'.

File	Commit Message	Time Ago
.github	Fix bug report template	4 days ago
Docs	doc generation	4 days ago
Licenses	Update license-GPL.txt	9 years ago
installer	NoiseModellingSetup.exe does not require administrator ...	2 months ago
noisemodelling-emission	Set params to be optional, fix docstring	4 days ago
noisemodelling-jdbc	Fix logger	4 hours ago
noisemodelling-pathfinder	Refactoring, use the atmospheric table for period based ...	5 days ago
noisemodelling-propagation	Check the parsing of favorable settings for the Netherlan...	4 days ago
noisemodelling-scripts	Fix logger	4 hours ago
.gitignore	clean files	5 months ago
.readthedocs.yaml	Merge branch 'main' into 5.X	last year
CITATION.cff	Update CITATION.cff	3 months ago
CODE_OF_CONDUCT.md	Create CODE_OF_CONDUCT.md	7 years ago
CONTRIBUTING.md	Fix the license link in CONTRIBUTING.md	2 weeks ago
Dockerfile	install python3 in docker for generating the help html pa...	3 months ago
LICENSE	Create License	7 years ago
README.md	fix: rename WPS references to Blocks/Builder in docs	3 months ago
docker-compose.yml	Add Docker files and instructions	5 months ago
pom.xml	fix log timestamp management	5 hours ago

README Code of conduct Contributing GPL-3.0 license

CI passing License (GPLv3) docs passing release v6.0.0

NoiseModelling

NoiseModelling is a free and open-source Java library for producing environmental noise maps from local to

About

An open-source model for calculating noise maps.

noisemodelling.readthedocs.io/

acoustics crossos-eu gis
java modelling noise
noise-modelling noise-propagation
pathfinder pathfinding
propagation research
sound-propagation

Readme
GPL-3.0 license
Code of conduct
Contributing
Cite this repository
Activity
Custom properties
234 stars
48 watching
96 forks
Repository

Releases 34

Release 6.0.0 Latest
4 months ago

+ 33 releases

Deployments 175

github-pages 37 minutes ago

Packages 1

noisemodelling

Contributors 29

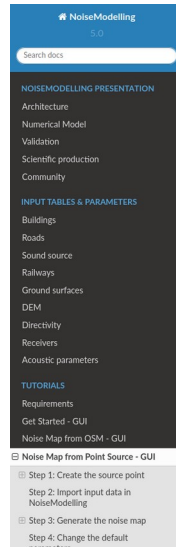
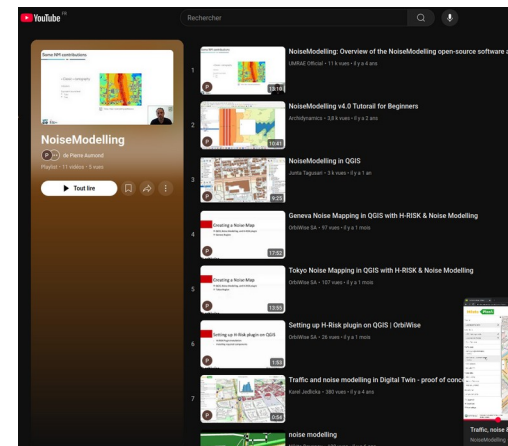
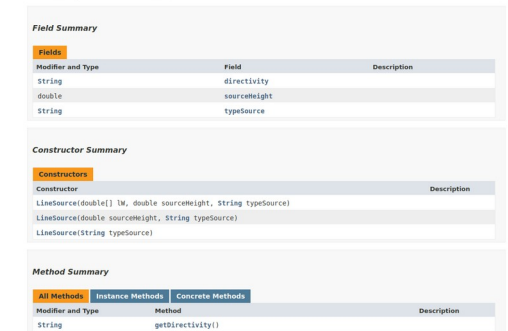
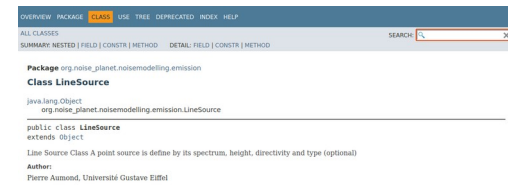
+ 15 contributors

Languages

Java 68.8% Groovy 27.7%
CSS 2.1% HTML 0.9%
JavaScript 0.4% NGIS 0.1%

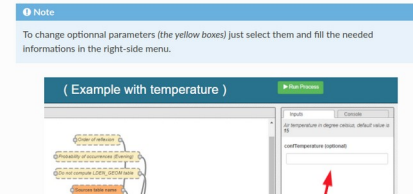
Website and documentations

Dev -
Javadoc



Step 4: Change the default parameters

To produce this noise map, we used, in most of WPS scripts, default parameters (e.g the height of the source, the number of reflections, the air temperature, ...). You are prompted to redo some of the previous steps by changing some of the settings. You will then be able to visually see what impact they have on the final noise map.



User doc

Videos -
demonstration
& tutorials

NoiseModelling website

NoiseModelling video presentation

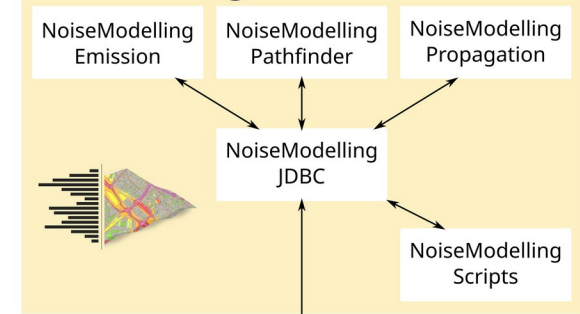


To see more videos about NoiseModelling, please have a look to this [dedicated playlist](#) on our YouTube channel.

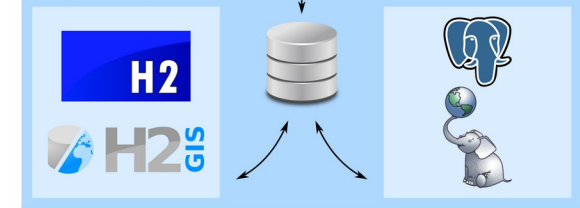
NoiseModelling core

- 5 Java libraries
 - Emission
 - Propa
 - Pathfinder
 - Database Connection
 - User scripts
- Direct connection to GIS databases
- User Interface or control via scripts (groovy, python, etc.)

NoiseModelling libraries



Database



Conformity to ISO 17534-1:2015

https://noisemodelling.readthedocs.io/en/latest/Cnossos_Report.html



INTERNATIONAL
STANDARD
ISO
17534-1:2015

Acoustics — Software for the
calculation of sound outdoors — Part
1: Quality requirements and quality
assurance

Read sample

ISO 17534-1:2015

Acoustics — Software for the calculation of sound outdoors

Part 1: Quality requirements and quality assurance

Published (Edition 1, 2015)

This publication was last reviewed and confirmed in 2020. Therefore this version remains current.

Test Case	Conform ?	NLD Conform ?	Largest Deviation	Details
TC01	☑	☑	0.00 dB @ 2000 Hz	TC01
TC02	☑	☑	0.07 dB @ 1000 Hz	TC02
TC03	☑	☑	0.05 dB @ 500 Hz	TC03
TC04	☑	☑	0.07 dB @ 1000 Hz	TC04
TC05	☑	☑	0.00 dB @ 1000 Hz	TC05
TC06	☑	☑	0.06 dB @ 500 Hz	TC06
TC07	☑	☑	0.02 dB @ 1000 Hz	TC07
TC08	☑	☑	0.02 dB @ 1000 Hz	TC08
TC09	☑	☑	0.01 dB @ 250 Hz	TC09
TC10	☑	☑	0.02 dB @ 250 Hz	TC10
TC11	☑	☑	0.02 dB @ 125 Hz	TC11
TC12	☑	☑	0.04 dB @ 500 Hz	TC12
TC13	☑	☑	0.01 dB @ 500 Hz	TC13
TC14	☑	☑	0.04 dB @ 8000 Hz	TC14
TC15	☑	☑	0.06 dB @ 250 Hz	TC15
TC16	☑	☑	0.00 dB @ 2000 Hz	TC16
TC17	☑	☑	0.05 dB @ 500 Hz	TC17
TC18	☑	☑	0.02 dB @ 500 Hz	TC18

Benchmark every version



RESULTS FOR VERSION	v4.0.0	v4.0.1	v4.0.2	v4.0.4	v4.0.5
Mean LAeq	54.94 dB	54.94 dB	55.99 dB	55.82 dB	55.99 dB
Receiver Count	00-11-31	00-10-03	00-10-04	00-11-04	00-10-07
Receiver Time Per Receiver	01.02	01.11	00.00	01.00	01.10
Receiver Time Per Receiver	01	01	01	01	01
Receiver Count	4035161	5050900	4037401	5303017	5500900
Receiver Time Per Receiver	01.00	01.10	01.10	01.10	01.10
Receiver Time Per Receiver	01	01	01	01	01
Receiver Count	145	145	145	145	145
Receiver Time Per Receiver	11.0.30.1	11.0.30.1	11.0.30.1	11.0.30.1	11.0.30.1
Receiver Time Per Receiver	11.0.30.1	11.0.30.1	11.0.30.1	11.0.30.1	11.0.30.1
Receiver Count	v4-custom	v4-custom	v4-custom	v4-custom	v4-custom

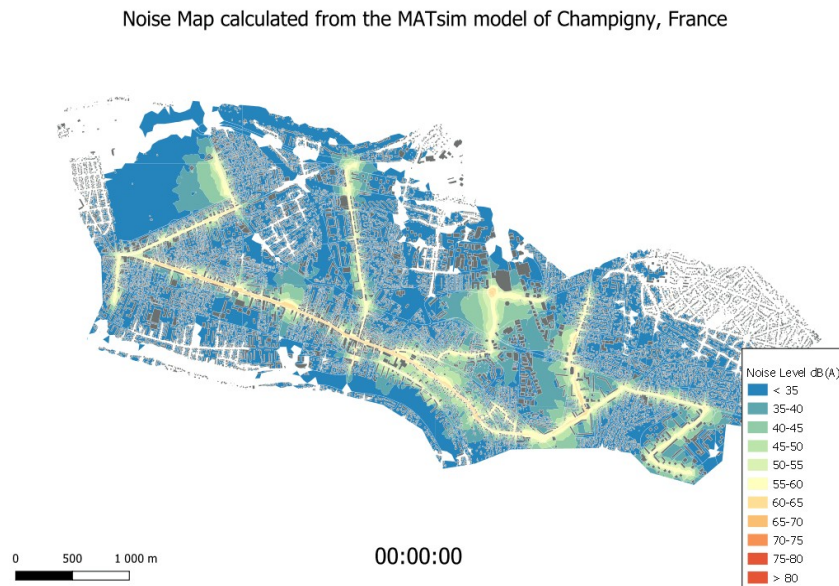
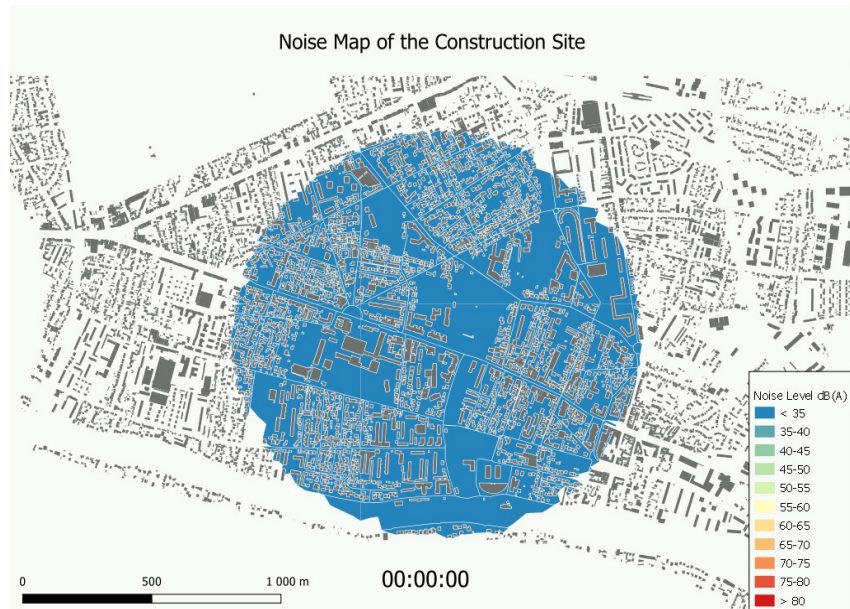
VERSION	REF.	OFFSET (DB)	MEAN ERROR	STD	RMSE	MAX ERROR	RECEIVERS
v4.0.0	#10	18.64	4.24	5.45	6.91	11.67	10/10
v4.0.1	#10	18.64	4.24	5.45	6.91	11.67	10/10
v4.0.2	#10	18.64	4.57	5.57	7.21	12.26	10/10
v4.0.4	#10	18.64	4.57	5.57	7.21	12.26	10/10
v4.0.5	#10	18.64	4.00	5.87	7.10	11.30	10/10
v5.0.0	#10	25.34	9.46	10.39	14.05	27.73	10/10
v5.0.1	#10	10.27	0.71	2.90	2.99	6.88	10/10
v6.0.0	#10	13.37	3.48	2.57	4.33	7.43	10/10
v6.0.2-SNAPSHOT	#10	9.16	1.14	4.39	4.54	11.39	10/10



What for ? Dynamic Noise Maps (flow traffic)



NM Days 2025



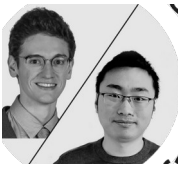
What for ? Data assimilation



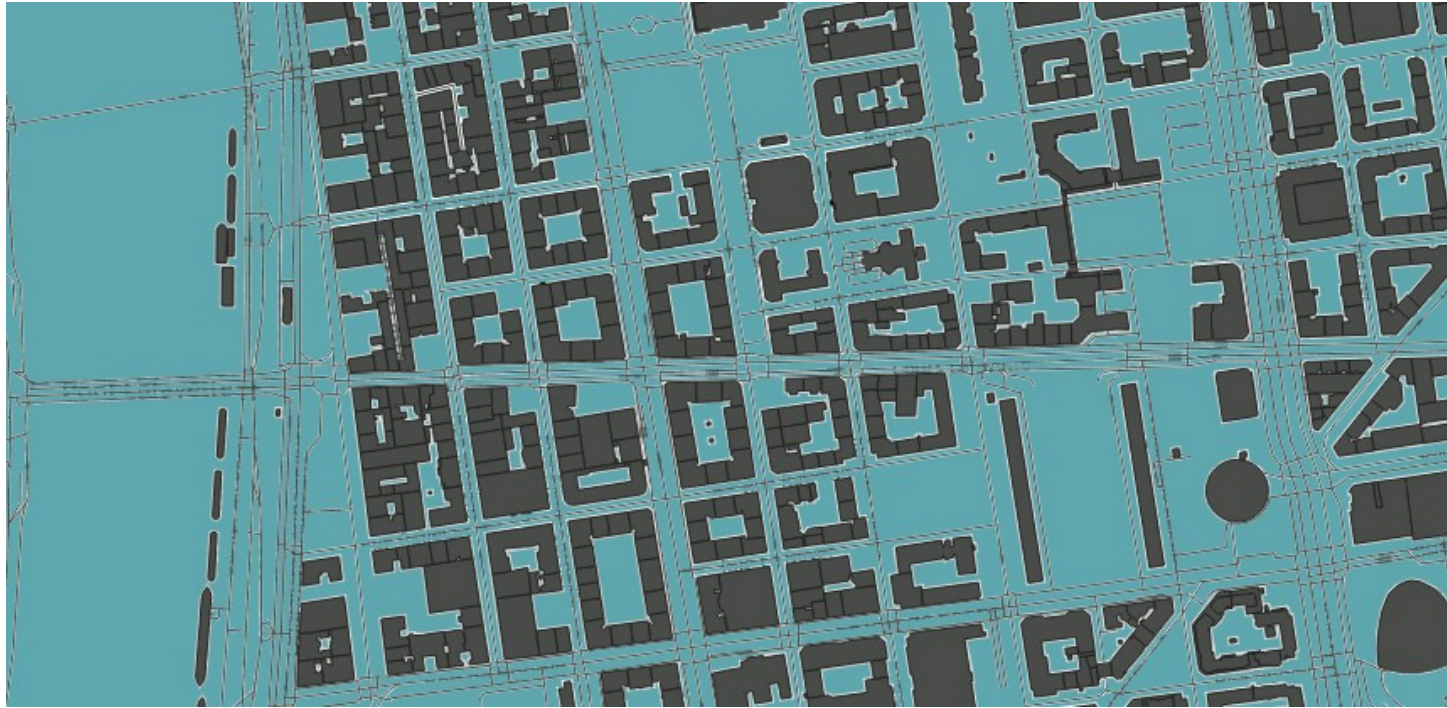
NM Days 2025



What for ? Dynamic Noise Maps (individualized traffic)



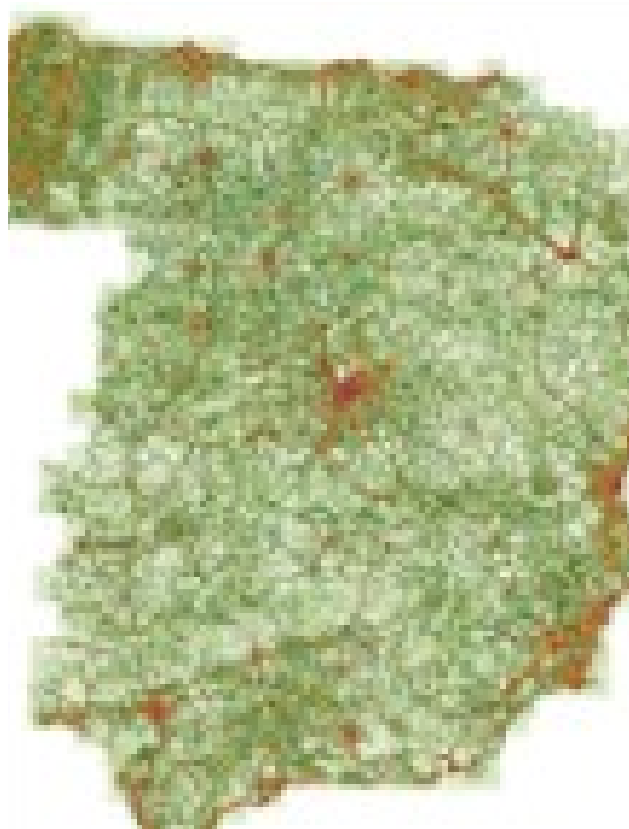
NM Days 2025



What for ? Noise Maps on HPC



Don't miss Ignacio's presentation to
view the unblurred image!

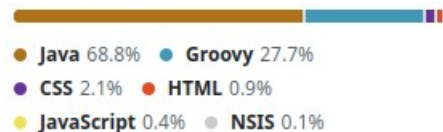


What for ?

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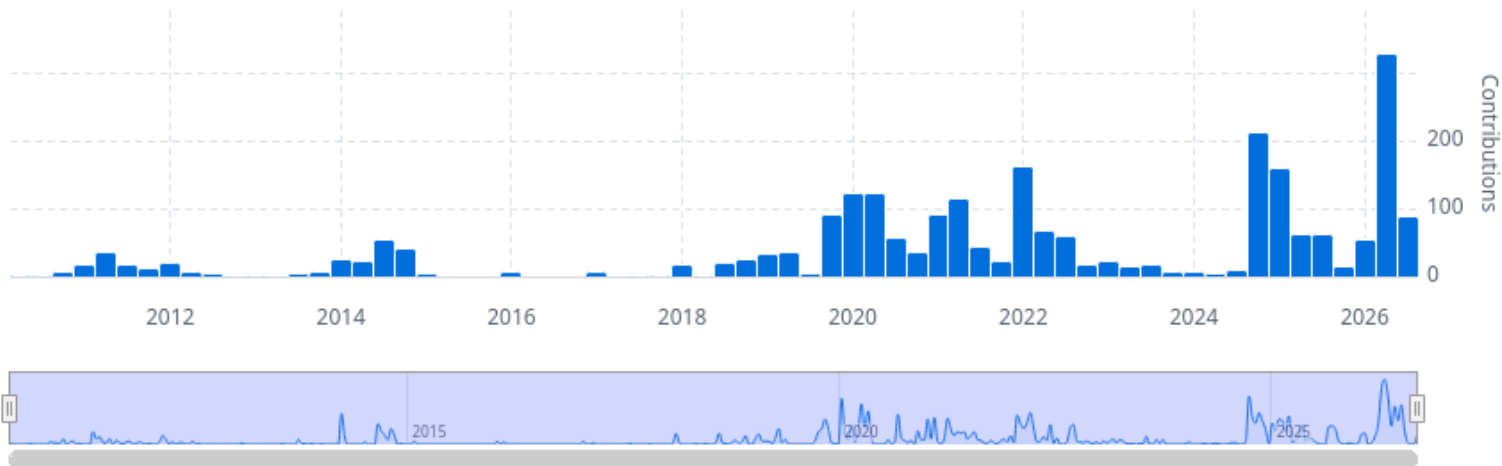
Some key figures

Languages



Commits over time

Weekly from 1 avr. 2010 to 20 sept. 2026



<https://github.com/Universite-Gustave-Eiffel/NoiseModelling/graphs/contributors>

Some key figures

- ~19k release downloads
- ~10 downloads/day
- 10 unique visits/day on GitHub page
- 234 GitHub stars
- 29 contributors



Universities & Research Institutes :

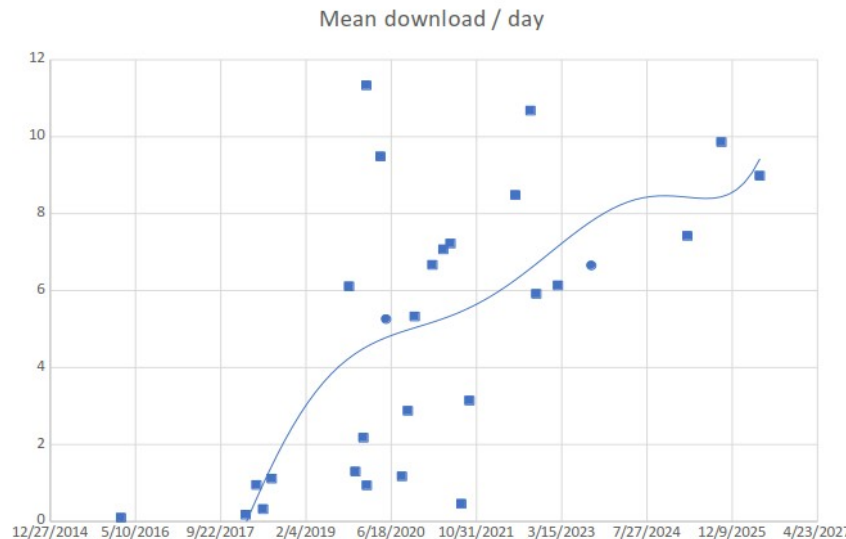
- KTH Royal, Utrecht University, Université de Sherbrooke (Canada), Groupe WAVES (Université de Gand - Belgique), Dipartimento di Ingegneria Civile/DICIV (Università di Salerne - Italie), HFT Stuttgart University of Applied Science, ONERA...

5 private companies

- Neovya, Quiet Places Ltd, Airbus, Orbiwise & UrbanMetrix

Public institutes :

- Centre National d'Incendie et de Secours - CNIS (Luxembourg), CEDEX, CEREMA...



Scientific production page

→ https://noisemodelling.readthedocs.io/en/latest/Scientific_production.html

Important to get numbers

numbers = visibility

visibility = interest

interest = fundings

fundings = development

dev = research

research = more fun :)

New visibility

Tell us if some papers are missing!

All	UMRAE	Not UMRAE	Noise Mapping (END)	Traffic dynamics
Other than Traffic noise		Fauna impacts	AI & Surrogates	Data Assimilation
Teaching		Software		

1. Cristian-Gabriel Alionte and Daniel-Constantin Comeaga. Noise assessment of the small-scale wind farm. *E3S Web Conf.*, 112:02011, 2019. Publisher: EDP Sciences. URL: https://www.e3s-conferences.org/articles/e3sconf/abs/2019/38/e3sconf_te-re-rd18_02011/e3sconf_te-re-rd18_02011.html (visited on 2025-06-06), doi:10.1051/e3sconf/201911202011.
2. Pierre Aumond, Erwan Bocher, David Ecotiere, Nicolas Fortin, Benoit Gauvreau, Gwenael Guillaume, and Gwendall Petit. Improvement of city noise map production processes and sensitivity analysis to noise models inputs. In *EuroNoise 2021 : 12th European Congress and Exposition on Noise Control Engineering*, 10 p. MADERE, Portugal, October 2021. URL: <https://hal.science/hal-03616731> (visited on 2025-06-10).
3. Pierre Aumond, Arna Can, Vivien Mallet, Benoit Gauvreau, and Gwenael Guillaume. Global

All	UMRAE	Not UMRAE	Noise Mapping (END)	Traffic dynamics
Other than Traffic noise		Fauna impacts	AI & Surrogates	Data Assimilation
Teaching		Software		

1. Antoine Lesieur, Pierre Aumond, Arnaud Can, and Vivien Mallet. Une cartographie du bruit plus rapide et plus précise combinant méta-modélisation et assimilation de données. In *16ème Congrès Français d'Acoustique, CFA2022*. Marseille, France, April 2022. Société Française d'Acoustique and Laboratoire de Mécanique et d'Acoustique. URL: <https://hal.science/hal-03848396> (visited on 2025-06-06).
2. Antoine Lesieur, Vivien Mallet, Pierre Aumond, and Arnaud Can. Data assimilation for urban noise mapping with a meta-model. *Applied Acoustics*, 178:107938, 2021. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0003682X21000311>, doi:10.1016/j.apacoust.2021.107938.
3. Antoine Lesieur, Vivien Mallet, Pierre Aumond, and Arnaud Can. Faster and more accurate noise mapping combining meta-modelling and data assimilation. In *Proceedings of Euronoise 2021*. Madeira, Portugal, 2021. URL: https://cense.ifsttar.fr/fileadmin/contributeurs/CENSE/Documents/Euronoise_2021_WP4.pdf.

23rd International
lications.rwth-

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dall Petit, and Judicaël
ional database with the
! Congress and Conference
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je entre la base de
3 pour la réalisation de
: (CFA2022). Marseille,

oiseModelling open-
! NOISE-CON Congress
4. URL: <https://>

!ling framework for
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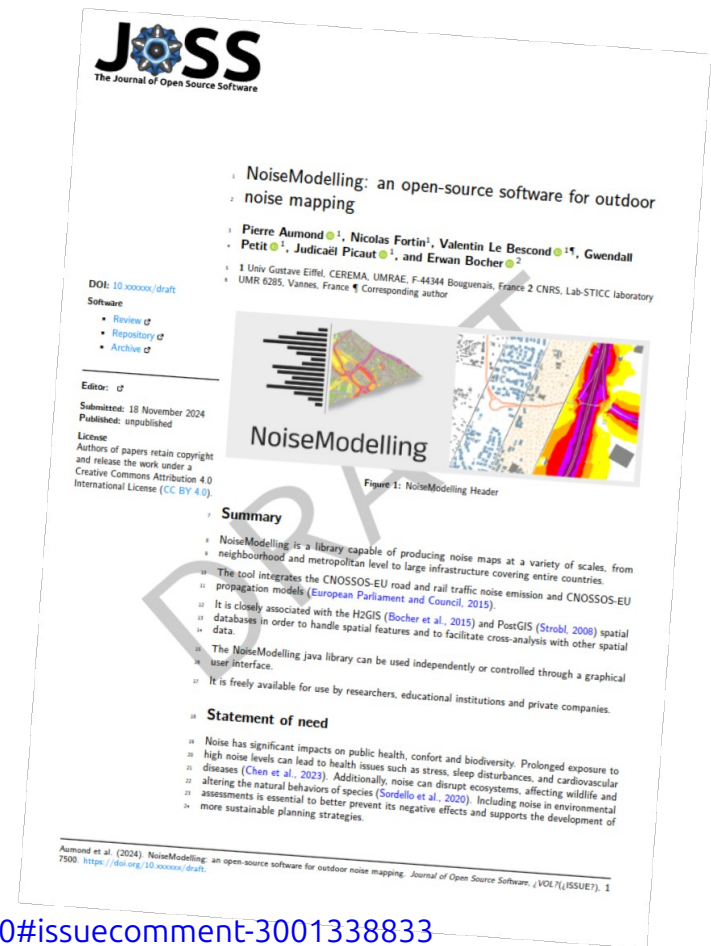
JOSS publication

Title: *"NoiseModelling: an open-source software for outdoor noise mapping"*

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More details: <https://github.com/openjournals/joss-reviews/issues/7500#issuecomment-3001338833>